# What range of sounds can we hear?

## Aim

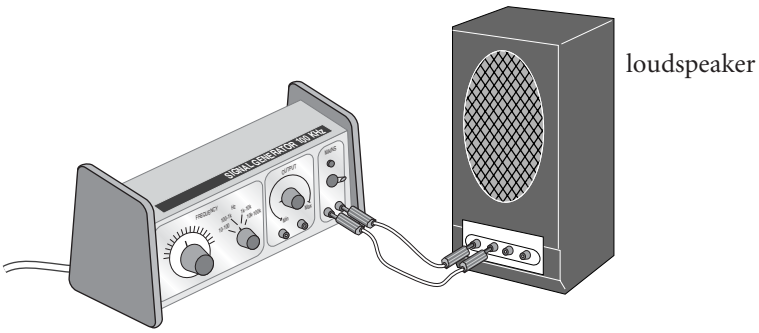
To find the highest- and lowest-pitched sounds you can hear.

## Equipment

- signal generator connected to a loudspeaker

## What to do

Your teacher will set up a signal generator with a speaker so that the electrical signal produces a sound which you can hear.




1. Listen carefully as the pitch is made lower. The number of vibrations each second is reduced so that the sound has a lower frequency or pitch. Eventually you will no longer be able to hear the sound.
2. Record the frequency (pitch) at which you cannot hear the sound any more.
3. Your teacher will tell you the frequency at which he or she cannot hear the sound any more. Record this frequency.
4. Listen carefully as the pitch is made higher by increasing the number of vibrations per second. It will become very high pitched and eventually you will no longer be able to hear it.
5. Record the frequency (pitch) at which you cannot hear the sound any more.
6. Your teacher will tell you the frequency at which he or she cannot hear the sound any more. Record this frequency.

(Remember that not everyone in the class will have the same results, just as if you all ran a race you would not all finish at the same time, or in a test you would not all get the same score.)

## Analyse

**1** Could the teacher hear higher-pitched sounds than most of the class? Or could most of the class hear higher-pitched sounds than the teacher?

**2** Who heard the lowest-pitched sounds, most of the class or the teacher?

**3** Do your results suggest a link between hearing and age?



© C Tear, 2002, The Heinemann Science Scheme

# A model of the eardrum

## Aim

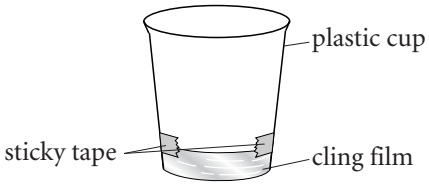
To make a model eardrum.

## Equipment

- plastic cup
- cling film
- sticky tape
- scissors
- mirror

## What to do

1. Using the scissors, make a hole in the bottom of the plastic cup. Now cut the bottom out of the cup.
2. Cut a piece of cling film large enough to completely cover the bottom of the cup. Stretch the cling film tightly over the bottom of the cup.
3. Use two or three pieces of sticky tape to stick the cling film to the sides of the cup.




4. Hold the cup over your mouth.
5. Look in the mirror so that you can see what is happening to the cling film as you try making these sounds:
   - **a** sssssssssssss
   - **b** hmmmmmmmm
   - **c** da da da da da da.

## Questions

**1** What happened to the cling film when you made each sound?

**2** Use your model to explain how the eardrum works.



© C Tear, 2002, The Heinemann Science Scheme

# Measuring sound levels

## Aim

To measure the loudness of different sounds.

## Equipment

- sound meter or sound sensor and data logger
- tape measure

## What to do

❶ Draw a table ready to record the type of sound you are testing (for example, students talking) and the level shown on the meter in decibels (in dB).

❷ Think of some sounds you can test around your school. List these in your table. In some cases you may need to say how far away you were from the sound.

❸ Measure and record the noise levels in your table.




## Analyse

**1** Draw a bar chart of your results.

**2** What was the loudest sound?

## Evaluate

**3** Levels of about 80 dB and higher are dangerous. Was this level reached during your experiment?

**4** Did you notice any safety notices or ear protection being used in any areas?

**5** Without the help of the meter, would you have put the sounds in the same order of loudness using your ears?

**6** You may have thought some sounds were louder than others but found that they were not really. Can you explain this?



© C Tear, 2002, The Heinemann Science Scheme

# How good are ear defenders?

## Aim

To find out which ear defenders block out the most sound.

## Equipment

- source of sound
- several different types of ear defender
- tape measure or metre ruler
- blindfold

## What to do

❶ Draw a table ready to record your results. It needs to show the types of ear defender used and the maximum distance at which the sound could be heard.

❷ Decide who is going to wear the ear defenders. It must be the same person every time. Blindfold this person so that they are only listening, not looking at the source of the sound.

❸ First take a result without any ear defenders. Set the sound level so that the blindfolded person cannot hear the sound about 4 m away. Measure and record this distance.

❹ Put the first ear defenders on the blindfolded person. Move the sound source towards the person until they can just hear it. Record this distance and move the sound source back to the start position.

❺ Repeat step ❹ for all the types of ear defender.

## Analyse

**1** Which pair of ear defenders blocked the sound closest to the person?

## Evaluate

**2** If another person did the same experiment, would you expect:

**a** the distances measured to be the same?

**b** the overall result (the order of the ear defenders) to be the same? Explain your answer.

**3** What do you think would happen if you changed the pitch (frequency) of the noise, so that it was higher or lower?

❻ If you have time, you could test this out.



© C Tear, 2002, The Heinemann Science Scheme